

Automated minus reference counts on the gold standard: protocol change

frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square-root axes - every asserted gold row

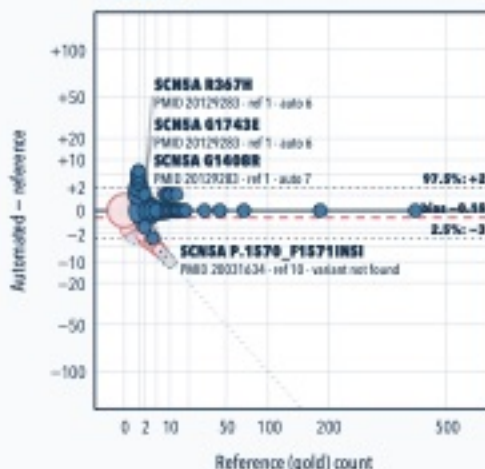
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

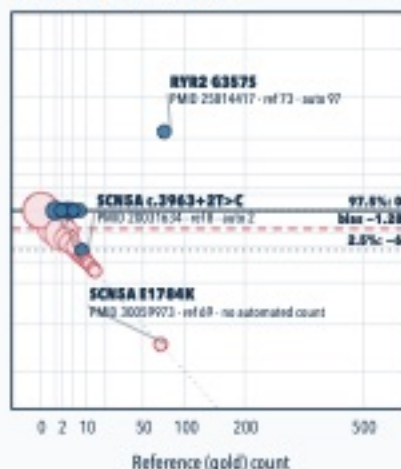
Baseline arm - 20260906_model12_grok43 - locked 2026-09-06 - 12 gene-paper attempts

Carriers



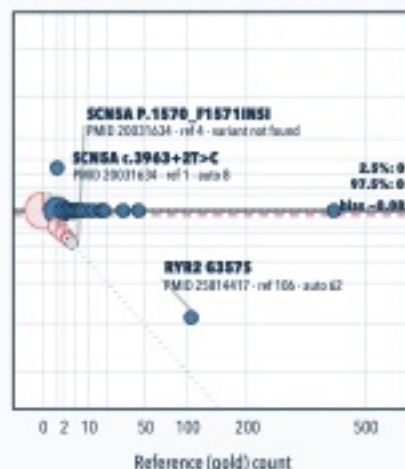
788 gold rows - matched 88.7% - supplied 70.2%
exact 74.5% (zero-baseline 18.5%) - within ± 1 92.5%
bias -0.19 - c2e MAE 0.428 - cond. MAE 0.179 (n 553)
empirical 95% limits -3 to +2 - RMSE 1.08

Affected individuals



788 gold rows - matched 88.7% - supplied 3.0%
exact 28.3% (zero-baseline 17.5%) - within ± 1 81.7%
bias -1.28 - c2e MAE 1.338 - cond. MAE 1.250 (n 24)
empirical 95% limits -6 to 0 - RMSE 3.22

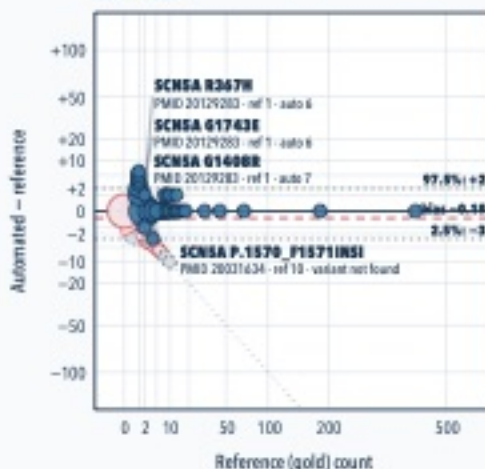
Unaffected individuals



787 gold rows - matched 88.7% - supplied 8.8%
exact 97.6% (zero-baseline 89.1%) - within ± 1 98.9%
bias -0.08 - c2e MAE 0.099 - cond. MAE 0.739 (n 69)
empirical 95% limits 0 to 0 - RMSE 1.61

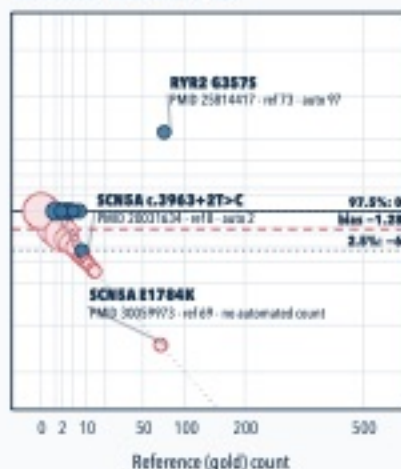
Candidate arm - 20260906_model12_grok43_astra_clinical_verified - locked 2026-09-06 - 12 gene-paper attempts

Carriers



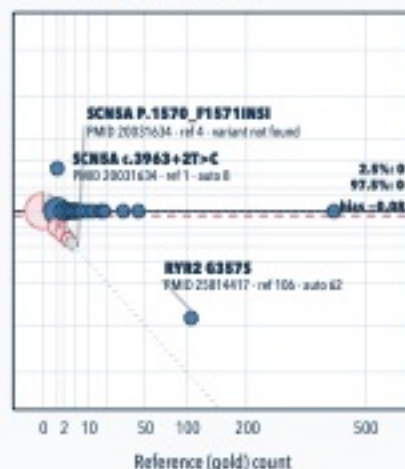
788 gold rows - matched 88.7% - supplied 70.3%
exact 74.6% (zero-baseline 18.5%) - within ± 1 92.6%
bias -0.19 - c2e MAE 0.435 - cond. MAE 0.179 (n 554)
empirical 95% limits -3 to +2 - RMSE 1.08

Affected individuals



788 gold rows - matched 88.7% - supplied 3.0%
exact 28.3% (zero-baseline 17.5%) - within ± 1 81.7%
bias -1.28 - c2e MAE 1.338 - cond. MAE 1.250 (n 24)
empirical 95% limits -6 to 0 - RMSE 3.22

Unaffected individuals



787 gold rows - matched 88.7% - supplied 8.8%
exact 97.6% (zero-baseline 89.1%) - within ± 1 98.9%
bias -0.08 - c2e MAE 0.099 - cond. MAE 0.739 (n 69)
empirical 95% limits 0 to 0 - RMSE 1.61

Candidate - baseline on the same 2363 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 0.428 $\rightarrow$ 0.425 (± 0.003)
supplied 553 $\rightarrow$ 554 - exact 587 $\rightarrow$ 589 - bias -0.19 $\rightarrow$ -0.19
rows better 1 - worse 0 - unchanged 787
coverage on matched 79.7% $\rightarrow$ 79.3% (± 0.14 pp)

Affected individuals

c2e MAE 1.338 $\rightarrow$ 1.338 (± 0.000)
supplied 24 $\rightarrow$ 24 - exact 160 $\rightarrow$ 160 - bias -1.28 $\rightarrow$ -1.28
rows better 0 - worse 0 - unchanged 788
coverage on matched 3.4% $\rightarrow$ 3.4% (± 0.00 pp)

Unaffected individuals

c2e MAE 0.099 $\rightarrow$ 0.099 (± 0.000)
supplied 69 $\rightarrow$ 69 - exact 768 $\rightarrow$ 768 - bias -0.08 $\rightarrow$ -0.08
rows better 0 - worse 0 - unchanged 787
coverage on matched 9.9% $\rightarrow$ 9.9% (± 0.00 pp)

Identity (primary endpoint): recall 88.71% $\rightarrow$ 88.71% (± 0.00 pp) - precision 95.10% $\rightarrow$ 95.10% (± 0.00 pp) - no registered comparison found; bounds not shown